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Transformation-Invariance Properties of Grad-CAM in Convolutional Neural Networks

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Introduction

Overview

Grad-CAM: The gradient-weighted class activation map (Grad CAM) produces a heat map that highlights important regions of an image using the target gradients of the final convolutional layer.

Transformation-Invariance: A method is transformation-invariant if its output remains consistent under input transformations (e.g., rotation, zoom, translation).

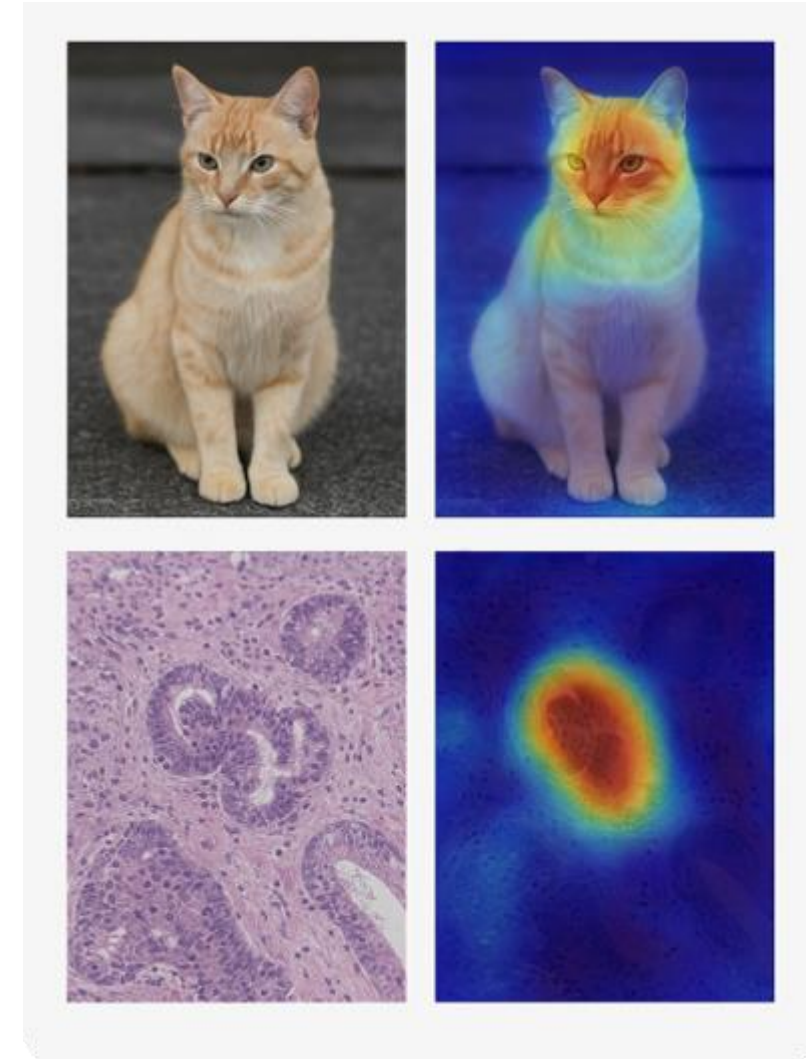


Fig 1: Grad-CAM Examples

Research Motivation

- **Core Issue:** Grad-CAM explanations are unstable under minor geometric transformations.
- **Real-world Context:** Real-world images naturally vary; explanations should remain stable to reflect consistent model reasoning.
- **Critical Challenges**
 - ❑ Reliability Crisis
 - ❑ Limited Generalization
 - ❑ Evaluation Gap
 - ❑ Missing Solutions
- **Research Goal:** Measure Grad-CAM's sensitivity and propose a realignment + averaging framework for transformation-invariant, reliable heatmaps.



Fig 2: Failure cases of Grad-CAM under transformations.

Research Objectives

- Quantify Grad-CAM sensitivity under rotation, zoom, and shifts
- Develop a pipeline to realign and average heatmaps
- Propose rotation/zoom/shift-invariant Grad-CAM variants
- Validate across ResNet152[1], DenseNet201[2], Xception[3]
- Assess with quantitative and qualitative analysis



Research Questions

RQ1: How sensitive is Grad-CAM to common spatial transforms?

RQ2: Can we reliably quantify sensitivity (L2, AUC-ROC)?

RQ3: Can inverse realignment + averaging improve invariance?

RQ4: Do architectures differ in robustness?

RQ5: Does the approach generalize to natural + medical images?



Related Work



Interpretability in Deep Learning

Two Approaches:

- **Intrinsic interpretability:** Simple, transparent models (e.g., decision trees, linear models) but often less accurate on complex tasks.[4]
- **Post-hoc interpretability:** Explain a trained, high-performing model (e.g., CNN) without modifying its architecture.[4]

This Work: Falls under post-hoc methods, explaining CNNs after training.

Grad-CAM and its Variant

Grad-CAM (Selvaraju et al., 2017) [5]:

- Uses gradients to compute class-specific importance in the final conv layer.
- Produces a class specific heatmap of important regions.

Variants: Grad-CAM++, Smooth Grad-CAM, Score-CAM, Ablation-CAM, etc.

- Improve sharpness or reduce noise.

Limitation: None explicitly address transformation invariance.

Issue: If an image is rotated or scaled, recomputed Grad-CAM may give a different map.

This Thesis: Quantifies Grad-CAM's instability under such transforms. Proposes a method to realign and average heatmaps for more stable explanations



Methodology

Proposed Architecture

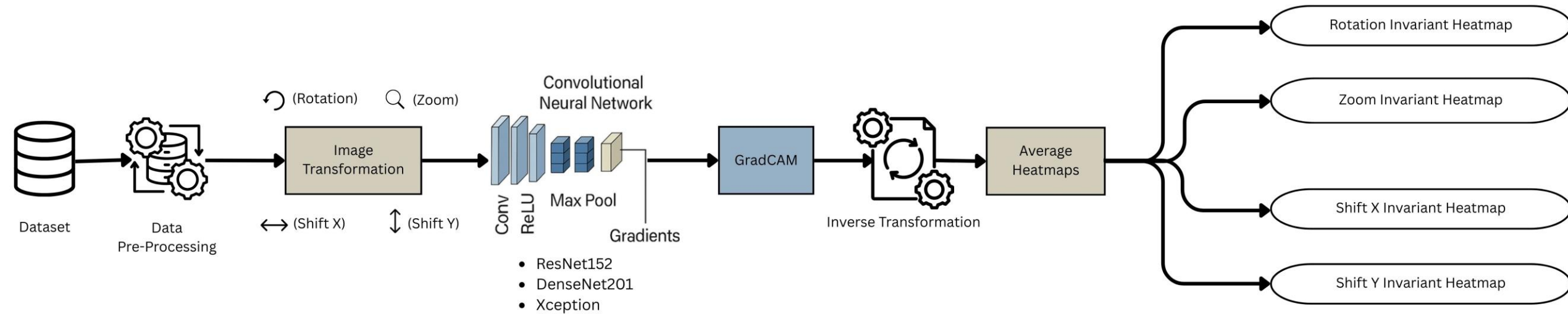


Figure 3: Overview of proposed methodology

Rotation Invariant Grad-CAM

To mitigate Grad-CAM's sensitivity to image orientation, we follow:

Forward Rotation:

For each angle $\theta_i \in [+180^\circ, -180^\circ]$ at 30° increments, rotate the original image I_{orig} to obtain

$$I_{rot}^i = Rotate(I_{orig}, \theta_i)$$

Grad-CAM on rotated inputs: Compute the Grad-CAM map on each I_{rot}^i .

$$H_{rot}^i = GradCAM(I_{rot}^i)$$

Inverse realignment: Rotate each heatmap back by $-\theta_i$ to the original frame.

$$H_{realign}^i = Rotate(H_{rot}^i, -\theta_i)$$

Averaging: Finally, we compute the pixel-wise average over these realigned maps to yield the rotation-invariant heatmap:

$$H_{rot-inv} = \frac{1}{N_\theta} \sum_{i=1}^{N_\theta} H_{realign}^i,$$

where N_θ = total number of rotation angles

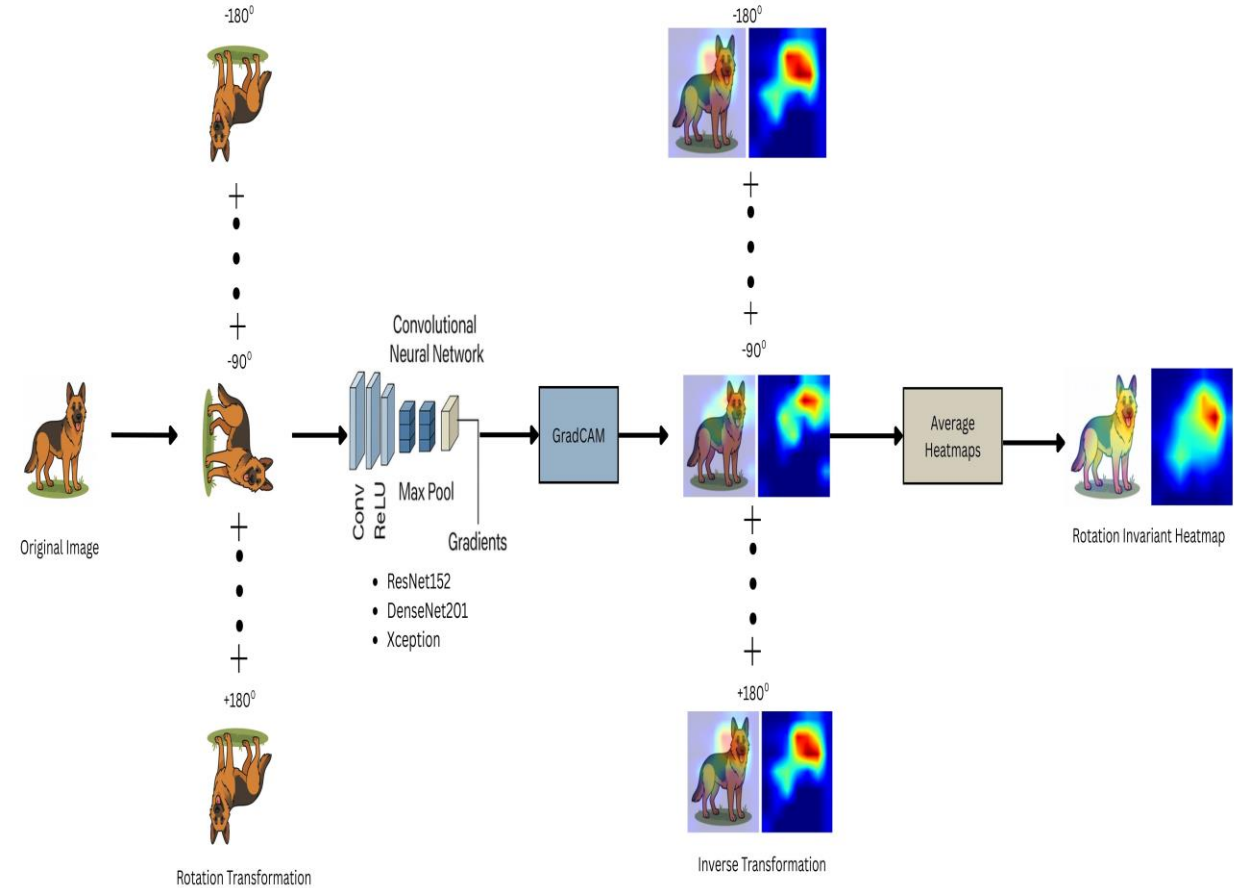


Fig 4: Methodology for Rotation Invariant Grad-CAM

Zoom Invariant Grad-CAM

Scale variations can cause Grad-CAM to focus on different spatial extents of an object. To compensate, we generate and combine heatmaps across multiple zoom levels:

Forward Zoom: For each zoom factor $Z_i \in \{0.67, 0.71, 0.77, 0.83, 0.91, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5\}$ resize the original image I_{orig} to obtain:

$$I_{zoom}^i = Zoom(I_{orig}, Z_i)$$

using bilinear interpolation and appropriate padding or cropping to maintain the field of view.

Grad-CAM on scaled images: Compute the Grad-CAM map on each I_{zoom}^i .

$$H_{Zoom}^i = GradCAM(I_{zoom}^i)$$

Restoring original resolution: Rescale each heatmap back to the original input dimensions:

$$H_{rescale}^i = Zoom(H_{Zoom}^i, orig_size)$$

Averaging: Finally, we compute the pixel-wise average over these realigned maps to yield the rotation-invariant heatmap:

$$H_{zoom-inv} = \frac{1}{N_z} \sum_{z=1}^{N_z} H_{rescale}^i ,$$

where $N_z = \text{total number of zoom factors}$

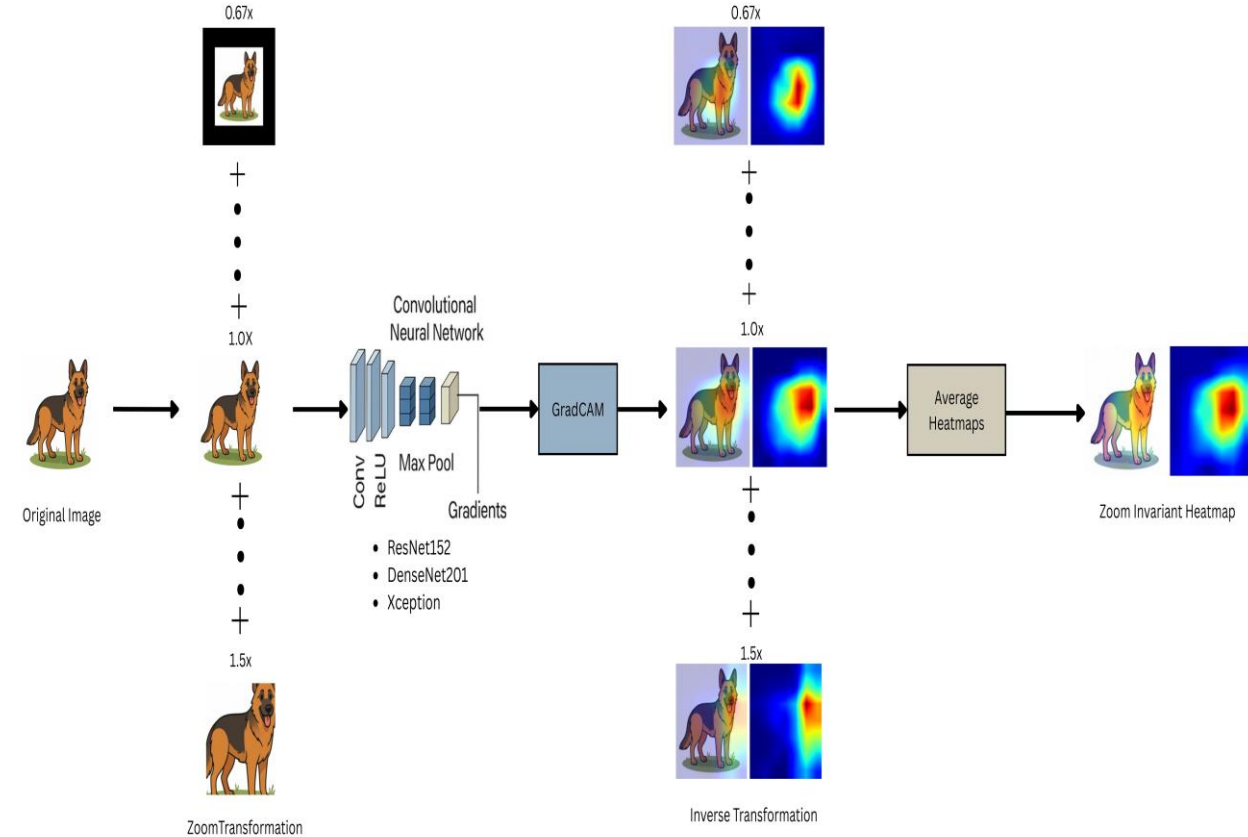


Fig 5: Methodology for Zoom Invariant Grad-CAM

Shift X Invariant Grad-CAM

Horizontal translations: sliding the image left or right—can shift the location of salient features and mislead a standard Grad-CAM map. To correct this, we follow:

Horizontal Shifting: For each axis, slide the image cyclically with every integer pixel $P_x \in [+180, -180]$ pixels, apply a cyclic (roll) horizontal shift to the original image I_{orig} to obtain:

$$I_{shift}^x = Shift_x(I_{orig}, P_x)$$

Grad-CAM on shifted images: Compute the Grad-CAM map on each I_{shift}^i .

$$H_{shift}^x = GradCAM(I_{shift}^x)$$

Inverse realignment: Shift each heatmap back by $-P_x$ to the original frame.

$$H_{realign}^x = Shift_x(H_{shift}^x, -P_x)$$

Averaging: Finally, we compute the pixel-wise average over these realigned maps to yield the shift-invariant X heatmap:

$$H_{shiftX-inv} = \frac{1}{N_x} \sum_{x=1}^{N_x} H_{realign}^x ,$$

where $N_x = \text{total number of shifted pixels to } x \text{ axis}$

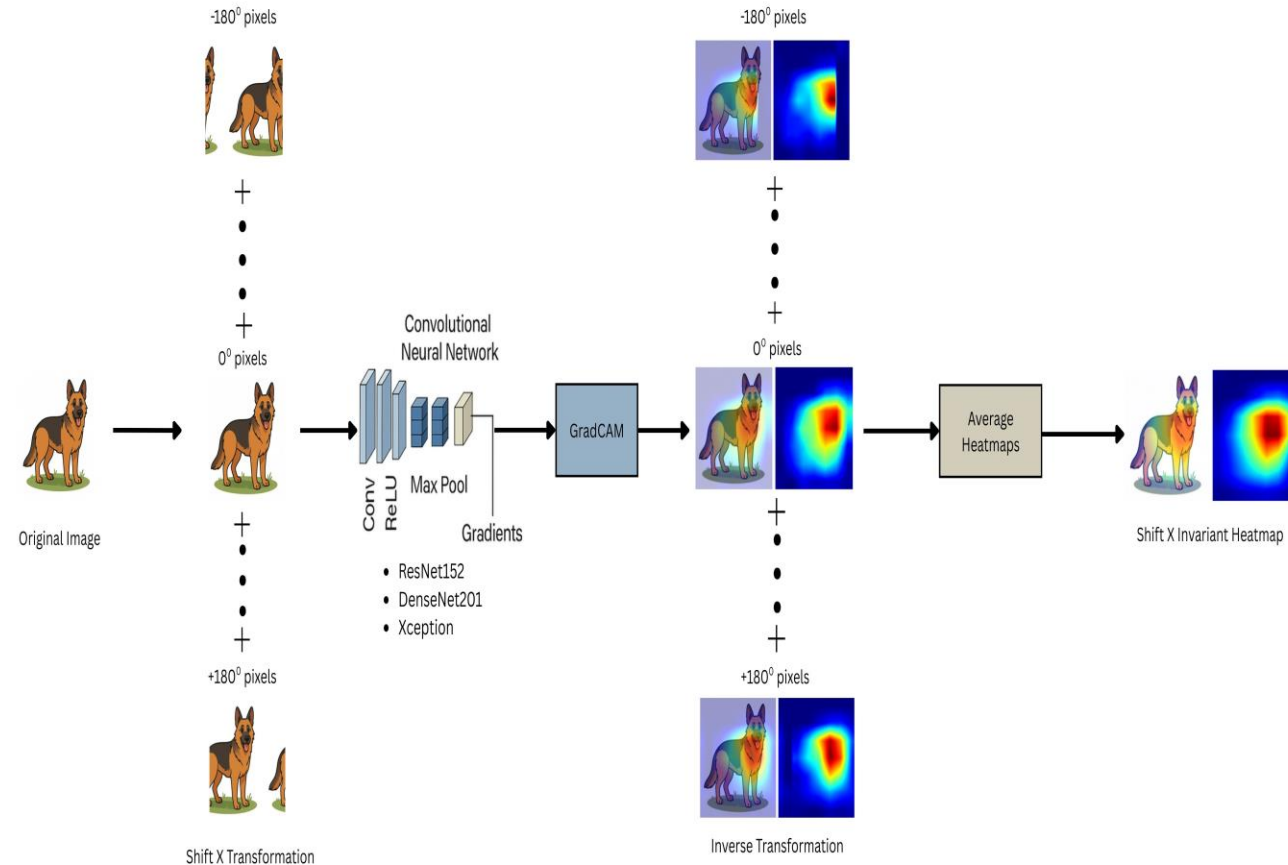


Fig 6: Methodology for Shift X Invariant Grad-CAM

Shift Y Invariant Grad-CAM

Similarly, vertical translations: moving the image up or down—can cause Grad-CAM to misplace its focus. We handle this:

Vertical Shifting: For each axis, slide the image cyclically by every integer pixel $P_y \in [+180, -180]$ pixels, apply a cyclic (roll) vertical shift to the original image I_{orig} to obtain:

$$I_{shift}^y = Shift_y(I_{orig}, P_y)$$

Grad-CAM on shifted images: Compute the Grad-CAM map on each I_{shift}^i .

$$H_{shift}^y = GradCAM(I_{shift}^y)$$

Inverse realignment: Shift each heatmap back by $-P_y$ to the original frame.

$$H_{realign}^y = Shift_y(H_{shift}^y, -P_y)$$

Averaging: Finally, we compute the pixel-wise average over these realigned maps to yield the shift-invariant Y heatmap:

$$H_{shiftY-inv} = \frac{1}{N_y} \sum_{y=1}^{N_y} H_{realign}^y,$$

where N_y = total number of shifted pixels to y axis

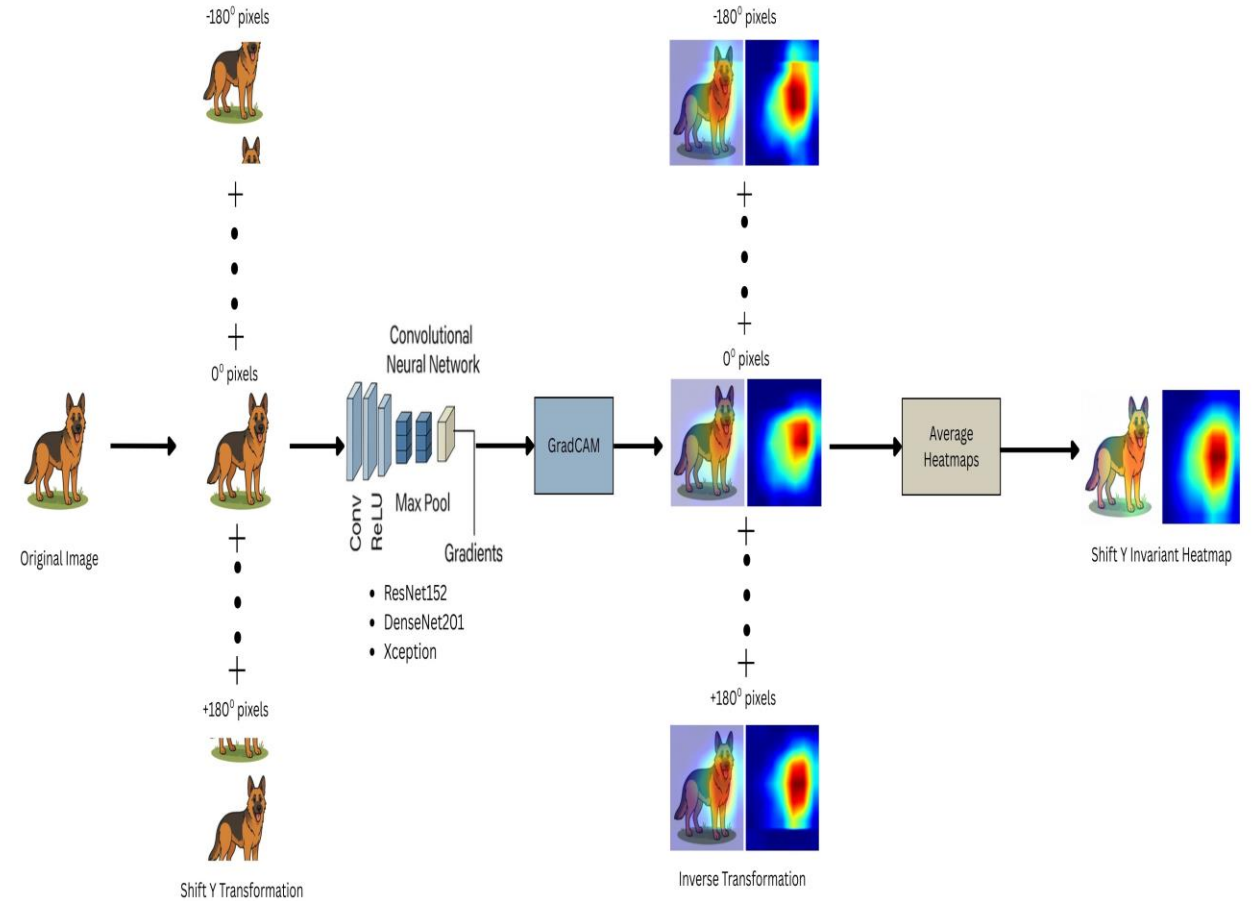


Fig 7: Methodology for Shift Y Invariant Grad-CAM



Experiments and Results

Experimental Setup

- Model Architectures

Model	Parameters (Millions)	Grad-CAM Layer	Input Size
ResNet152	60.2	conv5_block3_out	224×224
DenseNet201	20.2	conv5_block32_concat	224×224
Xception	22.9	block14_sepconv2_act	299×299

Table 1: Model architectures and Grad-CAM configuration

- Dataset

Dataset	Classes	Training	Validation	Test
MS COCO (2017)	80	118 287	5 000	20 000
CAMELYON16	2	64 000	8 000	8 000

Table 2: Dataset statistics for MS COCO (2017) [6] and CAMELYON16 patches [7]

Evaluation Criteria and Metrics

- **Goal:** Assess stability of Grad-CAM heatmaps under transformations.
- Two key approaches:
 - **Qualitative Evaluation** – Visual analysis for semantic consistency.
 - **Quantitative Evaluation** – Numerical metrics (L2, AUC).
- **Why both?** They offer complementary perspectives: visual interpretability + statistical analysis.

Qualitative Evaluation

- **Purpose:** Visually inspect if Grad-CAM focus remains consistent
- **Visual checks:**
 - Localization: does activation stay on class-relevant region?
 - Stability: does focus remain consistent across transformations?
 - Effect of averaging: does inverse alignment + averaging reduce “noise”?

Quantitative Evaluation – L2 Difference Metric

L2 Difference

- Measures pixel-wise difference between original and transformed heatmaps.
- Formula: Square root of the sum of squared pixel differences.

$$L2(H_{\text{original}}, H_{\text{transformed}}) = \sqrt{\sum_{i=1}^n (H_{\text{original}}^i - H_{\text{transformed}}^i)^2}$$

- Lower L2 = More similar = Better stability.
- Used in prior robustness studies: Kindermans (2019), Ghorbani (2019)

Quantitative Evaluation – AUC Metric

Area Under the ROC Curve (AUC)

- AUC measures how well a model separates positives from negatives across all decision thresholds.
- A higher AUC $\rightarrow$ stronger alignment and stability of visual explanations.

Using Grad-CAM heatmap scores:

- Normalize to $[0, 1]$ $\rightarrow$ this is the saliency map H with pixel scores $s_i \in [0,1]$.
- Compare H to the binary segmentation mask Y with labels $y_i \in [0,1]$. (1 = ROI, 0 = background)

Thresholding and ROC construction

For each pixel i with saliency score s_i , and threshold $\tau \in [0,1]$, form a binary prediction:

- $\hat{y}_i(\tau) = 1$ if $s_i \geq \tau$ predict as positive and $\hat{y}_i(\tau) = 0$ if $s_i < \tau$ predict as negative.

For each τ , compute:

- **TPR** (τ) = **TP** / (**TP** + **FN**) (true positive rate)
- **FPR** (τ) = **FP** / (**FP** + **TN**) (false positive rate)

Plot pairs (**TPR**, **FPR**) to get the ROC curve.

Quantitative Evaluation – AUC Metric

AUC Calculation:

- $AUC = \int_0^1 TPR(FPR) dFPR$

Interpretation: AUC = 1.0 means perfect separation; AUC = 0.5 means random guessing.

Aggregate Reporting:

Across multiple images or transformations:

- $AUC \pm SD = \mu_{AUC} \pm \sigma_{AUC}$

Where

- $\mu_{AUC} = \frac{1}{N} \sum AUC_k, \sigma_{AUC} = \sqrt{\frac{1}{N} \sum (AUC_k - \mu_{AUC})^2}$

Reports both **average discriminative power** and **robustness** under variations.

Experiment 1: Single-Image Transformation Consistency

Objective:

To assess how *Grad-CAM* explanations behave under different transformations on a single image, testing their **transformation-invariance**.

Setup:

- A single representative image was used as input.
- Applied four transformations:
 - Rotation
 - Zoom
 - Horizontal Shift (Shift X)
 - Vertical Shift (Shift Y)
- Model: ResNet152, DenseNet201, Xception
- Each transformed heatmap was inverse-realigned to the original coordinate frame.
- Final invariant representation was produced via pixel-wise averaging of all aligned heatmaps.

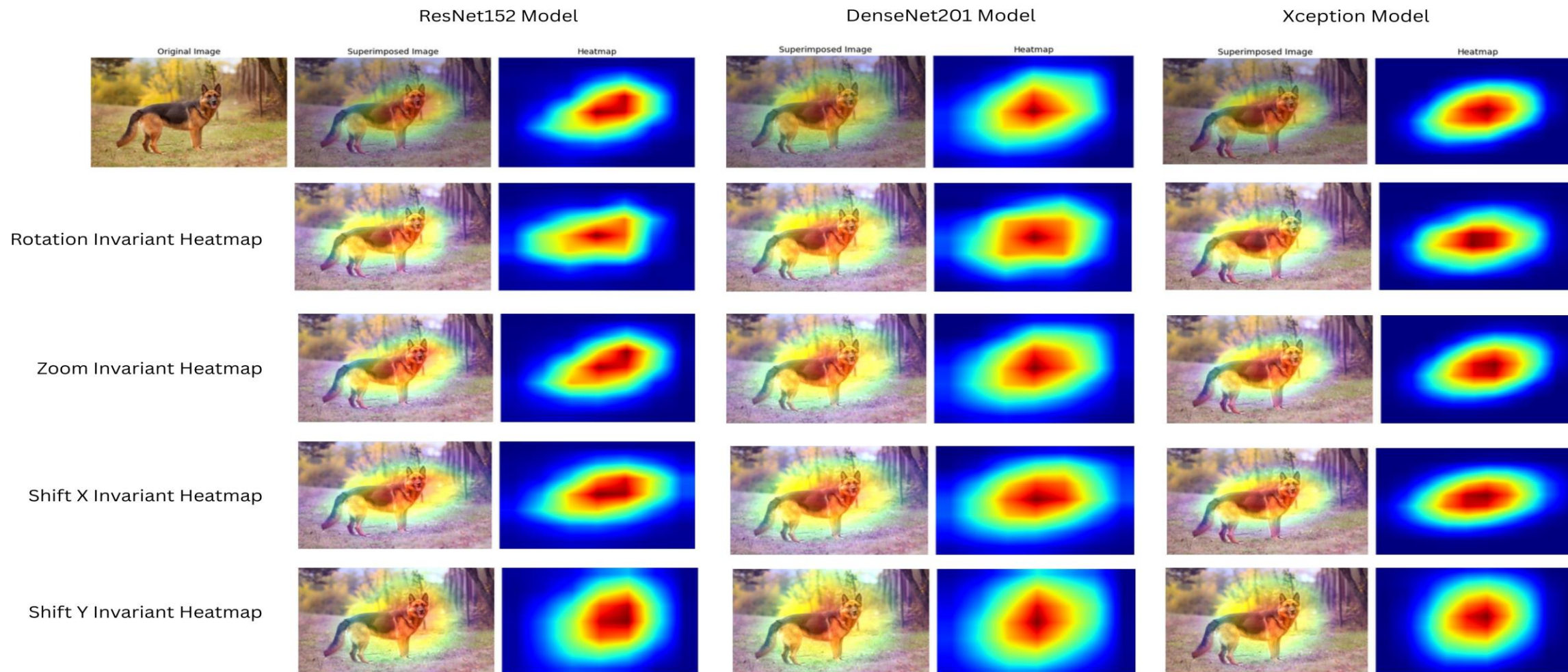
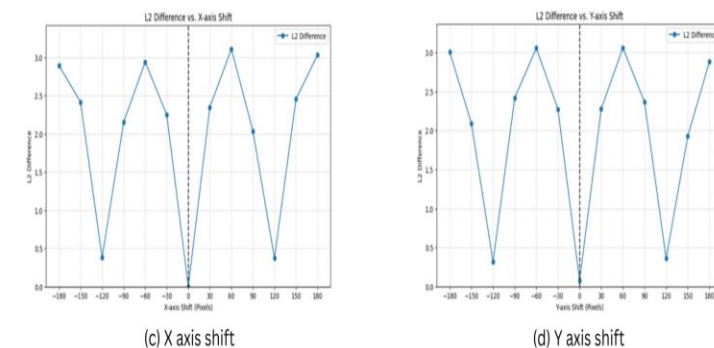
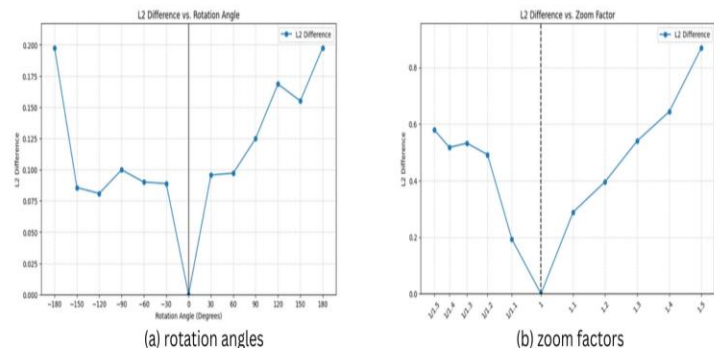
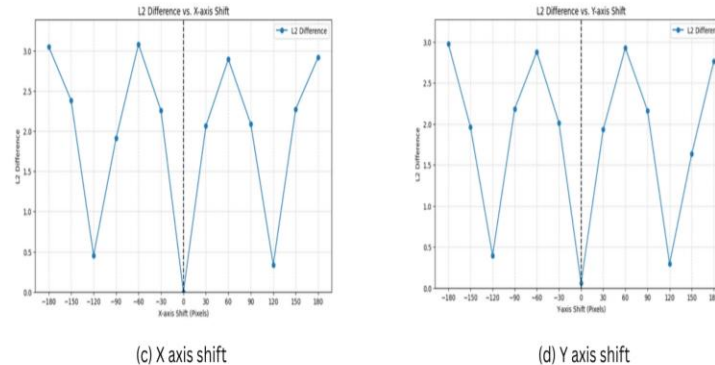
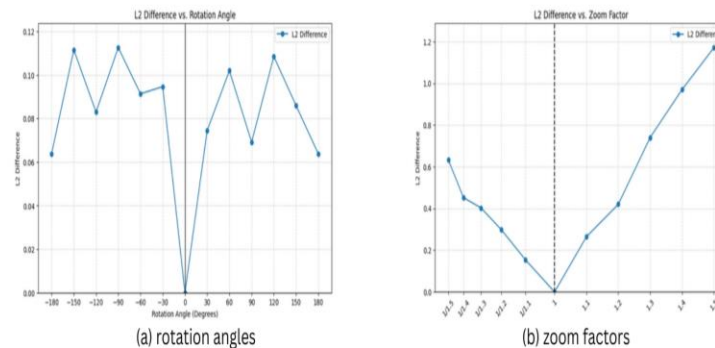


Figure 9: Grad-CAM activation heat-map visualization under transformations (rotation, zoom, shift X, shift Y) on a single image using ResNet152, DenseNet201, Xception

L2 Difference
(ResNet152)



L2 Difference
(DenseNet 201)



L2 Difference
(Xception)

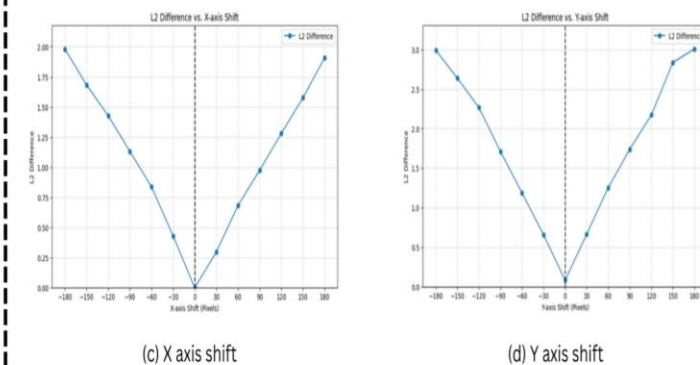
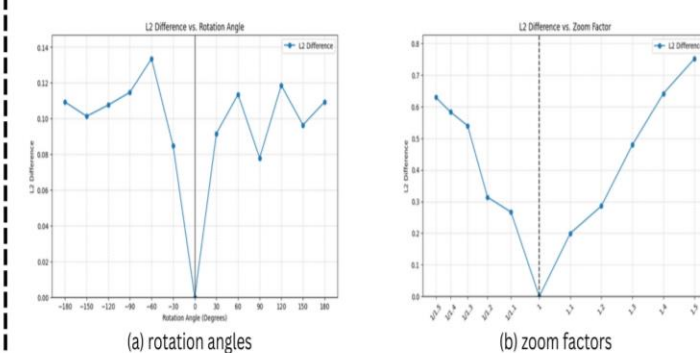


Figure 10: Quantifying Grad-CAM's sensitivity using L2 difference under varying (a) rotation angles, (b) zoom factors, (c) x-axis shifts, and (d) y-axis shifts for a single image using ResNet152, DenseNet201 and Xception

Experiment 2: Class-Specific Evaluation

Objective:

To analyze the class-specific transformation-invariance of Grad-CAM explanations for different object categories.

Setup:

- Targeted two object classes: “Dog” and “Cat.”
- Applied the same four transformations:
 - Rotation
 - Zoom
 - Horizontal Shift (Shift X)
 - Vertical Shift (Shift Y)
- Generated Grad-CAM heatmaps using three CNN architectures:
 - ResNet152
 - DenseNet201
 - Xception

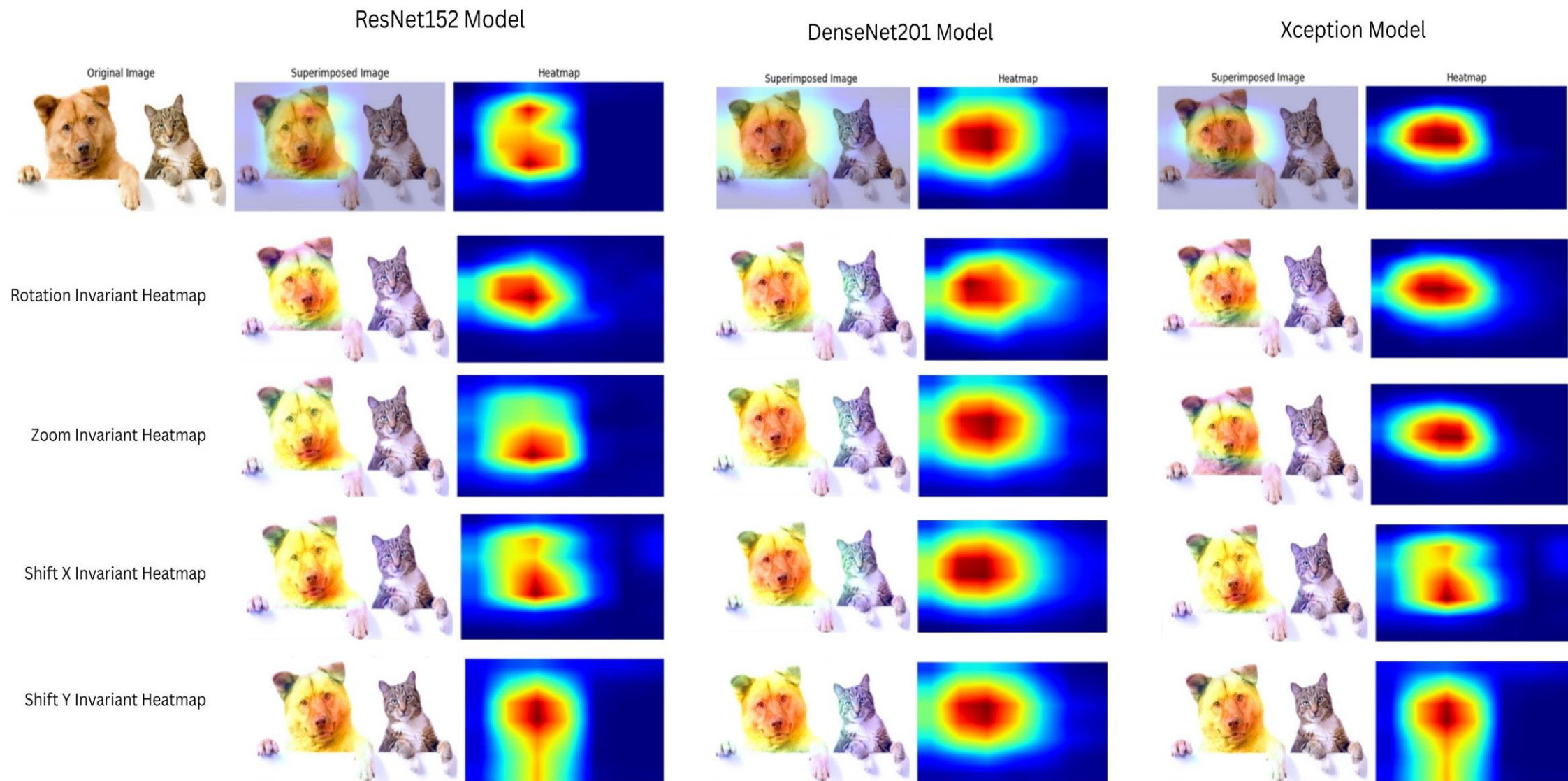
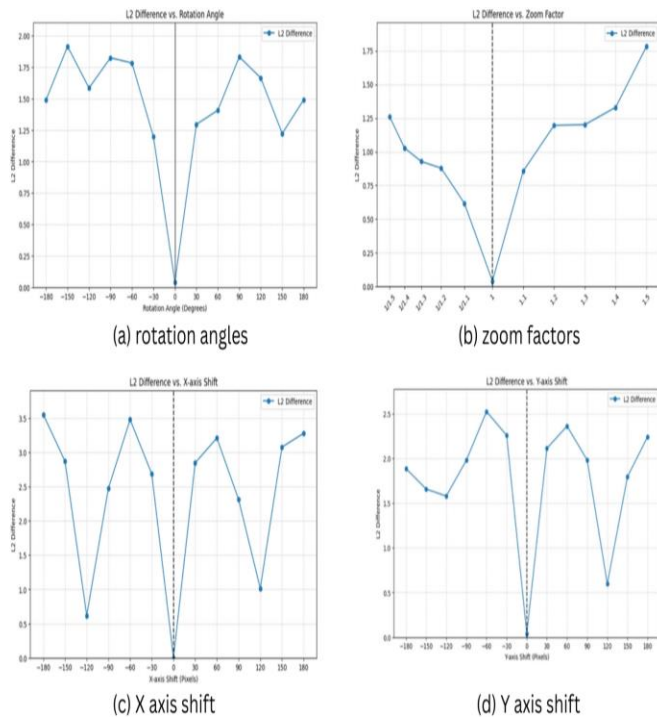
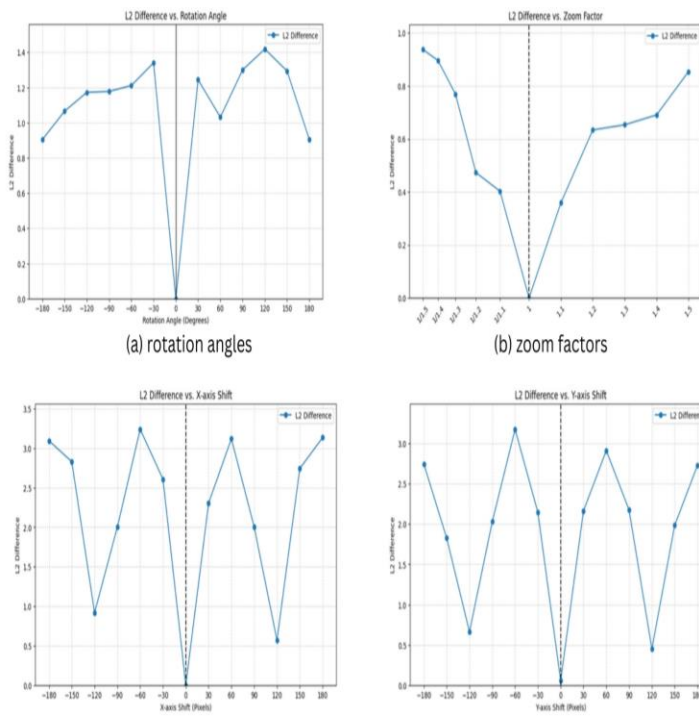


Figure 11: Transformation-invariant Grad-CAM heatmaps for "Dog" keyword class under basic transformations using ResNet152, DenseNet201 and Xception models

L2 Difference
(ResNet152)



L2 Difference
(DenseNet 201)



L2 Difference
(Xception)

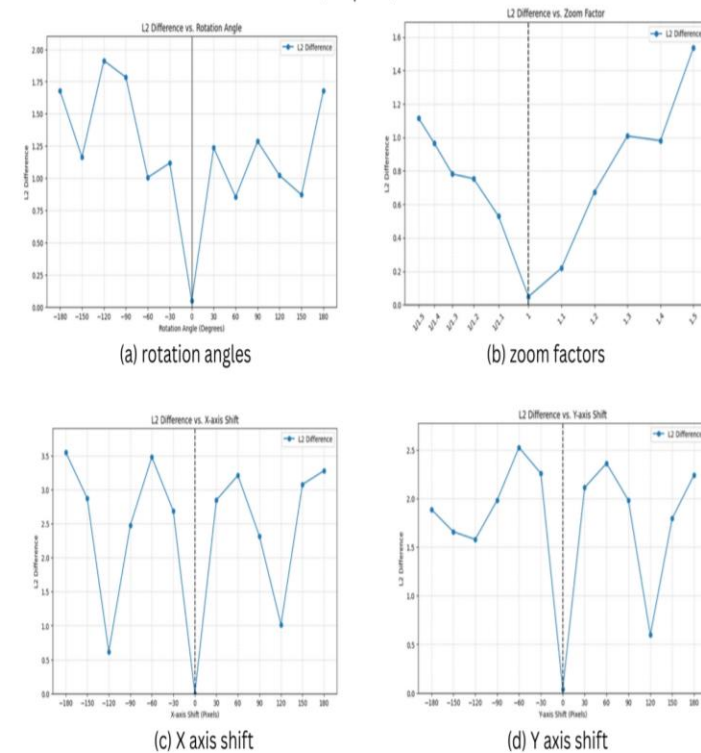


Figure 12: Quantifying Grad-CAM's sensitivity using L2 difference under varying (a) rotation angles, (b) zoom factors, (c) x-axis shifts, and (d) y-axis shifts for "Dog" keyword using ResNet152, DenseNet201 and Xception

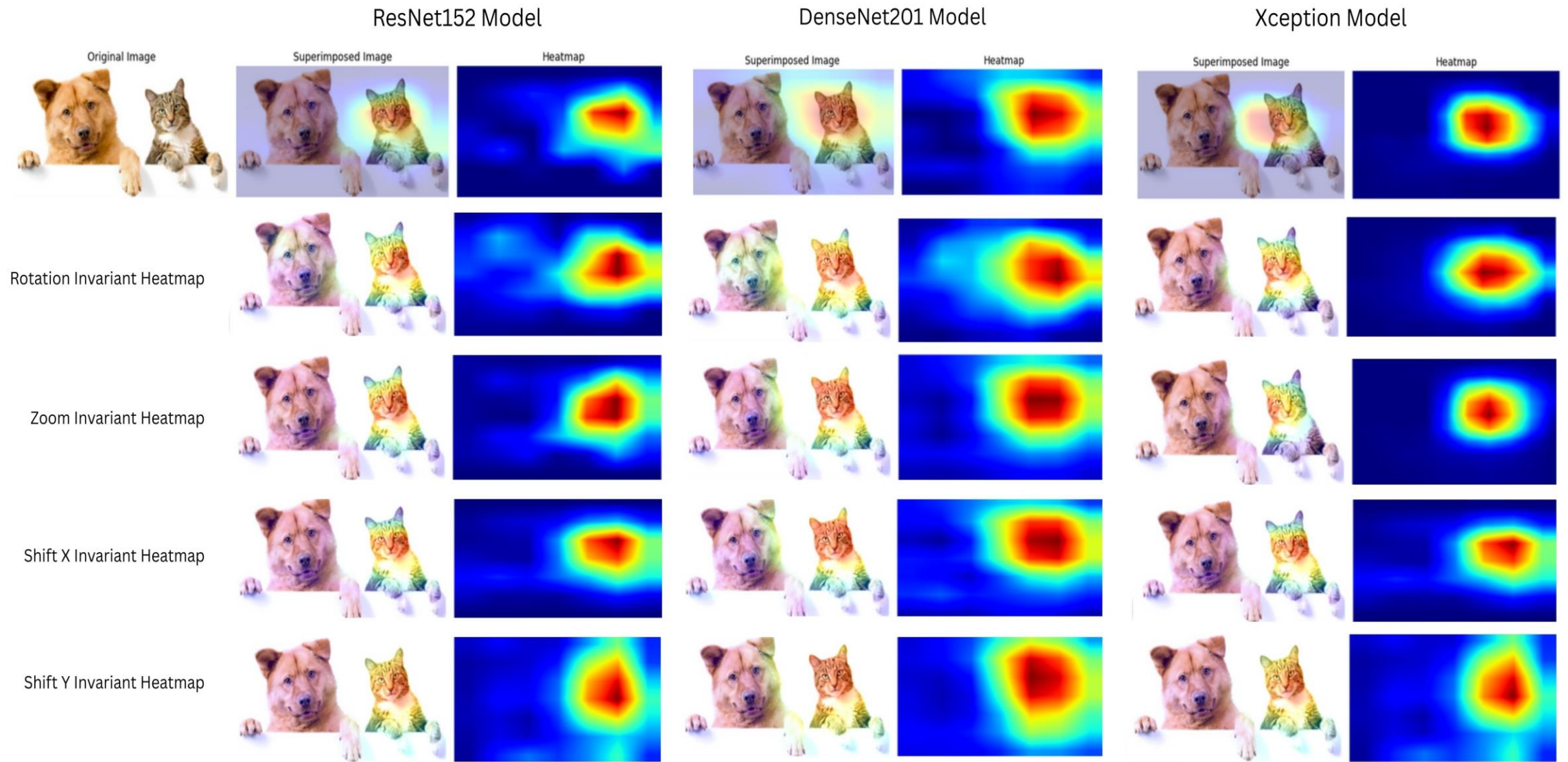
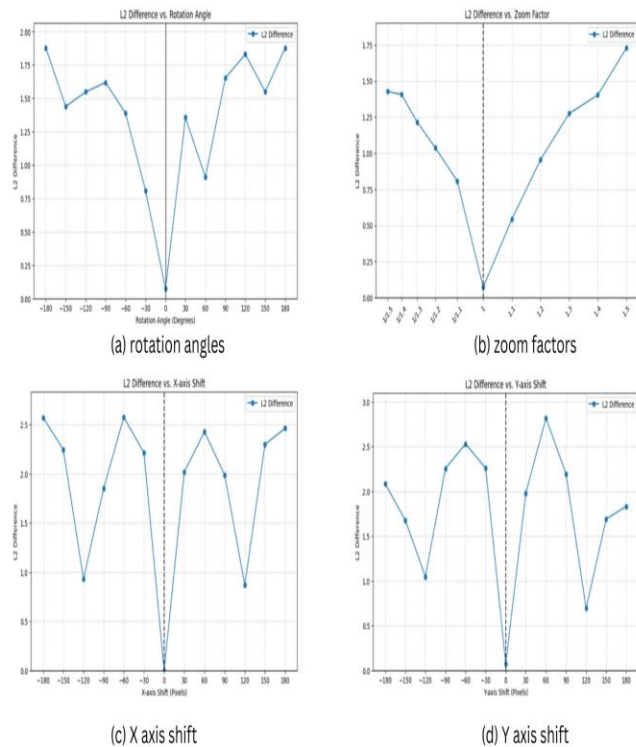
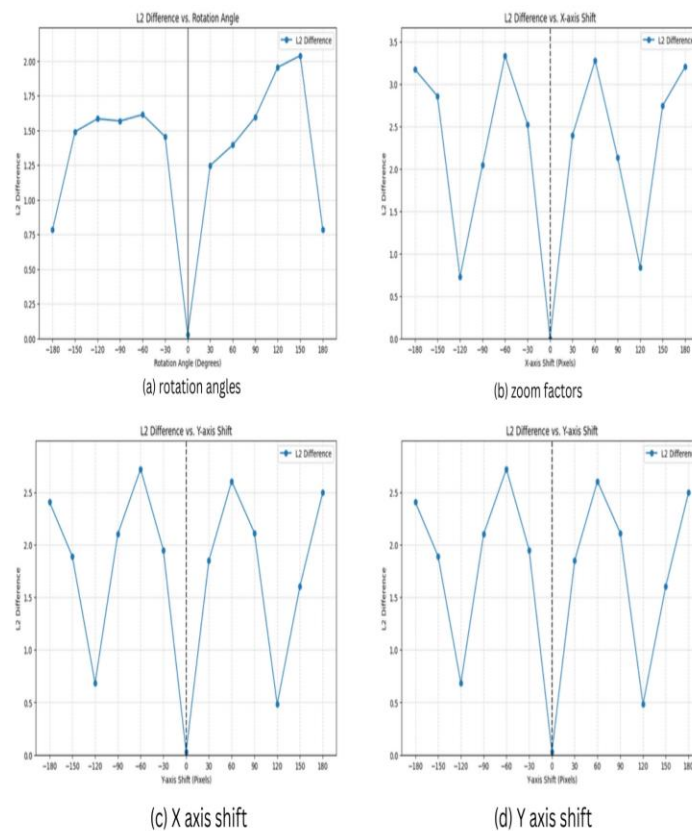


Figure 13: Transformation-invariant Grad-CAM heatmaps for "Cat" keyword class under basic transformations using ResNet152, DenseNet201 and Xception models

L2 Difference
(ResNet152)



L2 Difference
(DenseNet201)



L2 Difference
(Xception)

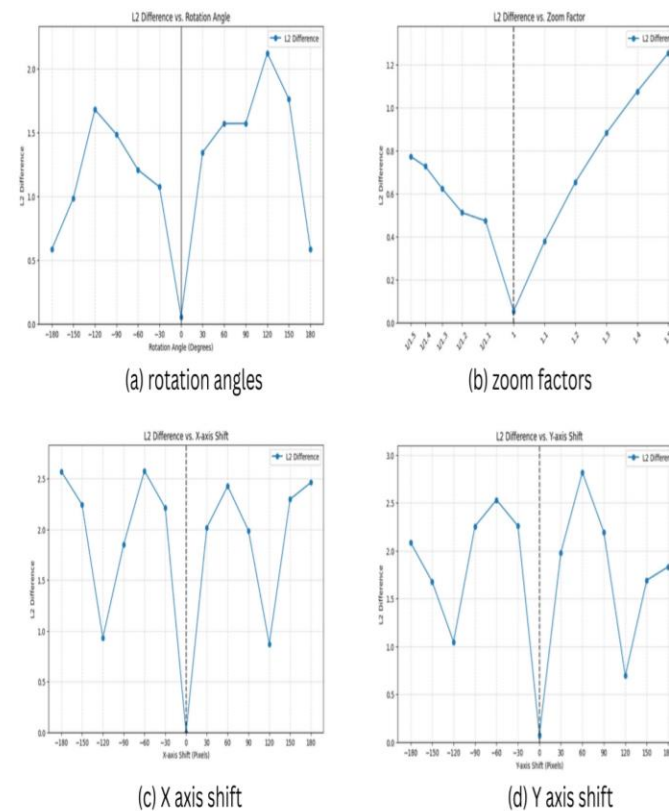


Figure 14: Quantifying Grad-CAM's sensitivity using L2 difference under varying (a) rotation angles, (b) zoom factors, (c) x-axis shifts, and (d) y-axis shifts for "Cat" keyword using ResNet152, DenseNet201 and Xception

Experiment 3: Dataset Level Evaluation

Objective:

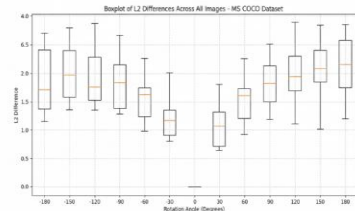
To evaluate the scalability and robustness of transformation-invariant Grad-CAM on a large-scale, real-world dataset (**MS COCO**) containing 80 object categories with pixel-level segmentation masks.

Setup:

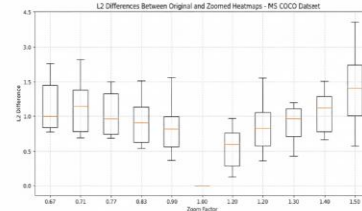
- Built binary ground-truth masks from **COCO polygons/RLE** annotations.
- Generated class-specific Grad-CAM heatmaps for each combination of:
 - Backbones: ResNet152, DenseNet201, Xception
 - Settings: Baseline, Rotation-invariant, Zoom-invariant, Shift-X-invariant, and Shift-Y-invariant
- Invariant heatmaps = transformed → inverse-aligned → averaged → normalized

Models	Baseline		Rotation Invariant Grad-CAM		Zoom Invariant Grad-CAM		Shift X Invariant Grad-CAM		Shift Y Invariant Grad-CAM	
	L_2	AUC ($\mu \pm \sigma$)	L_2	AUC ($\mu \pm \sigma$)	L_2	AUC ($\mu \pm \sigma$)	L_2	AUC ($\mu \pm \sigma$)	L_2	AUC ($\mu \pm \sigma$)
ResNet152	0	0.832 ± 0.021	0.25	0.851 ± 0.017	0.22	0.871 ± 0.015	0.18	0.881 ± 0.016	0.20	0.870 ± 0.017
DenseNet201	0	0.846 ± 0.019	0.21	0.874 ± 0.018	0.19	0.892 ± 0.013	0.15	0.901 ± 0.014	0.17	0.895 ± 0.011
Xception	0	0.827 ± 0.023	0.29	0.843 ± 0.019	0.26	0.862 ± 0.018	0.22	0.849 ± 0.020	0.23	0.844 ± 0.021

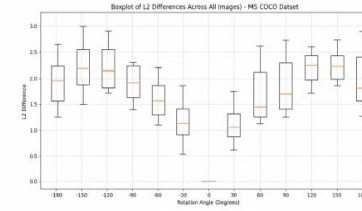
Table 3: MS Coco: Baseline vs. Transformation-Invariant Grad-CAM (L2 differences and Area Under the Curve)



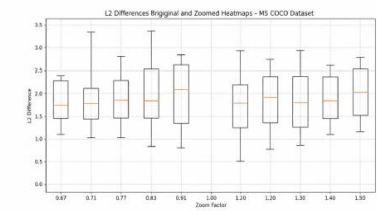
(a) L_2 differences of Rotation Invariant



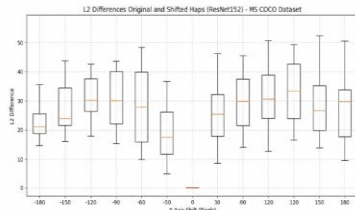
(b) L_2 differences of Zoom Invariant



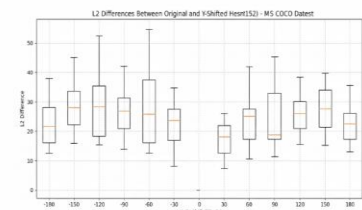
(a) L_2 differences of Rotation Invariant



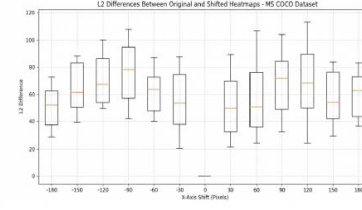
(b) L_2 differences of Zoom Invariant



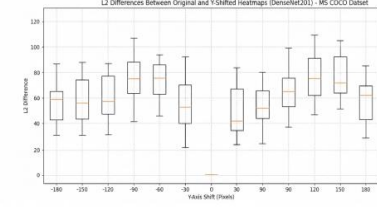
(c) L_2 differences of Shift X Invariant



(d) L_2 differences of Shift Y Invariant



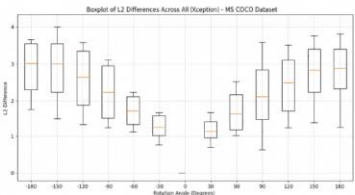
(c) L_2 differences of Shift X Invariant



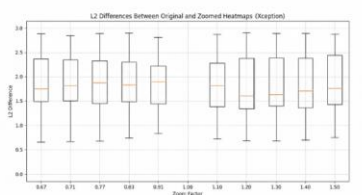
(d) L_2 differences of Shift Y Invariant

Figure 15: L_2 differences between Baseline and transformation-invariant Grad-CAM heatmaps for ResNet152

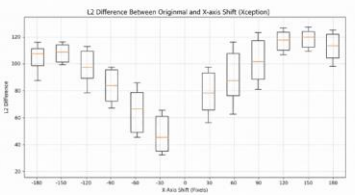
Figure 16: L_2 differences between Baseline and transformation-invariant Grad-CAM heatmaps for DenseNet201



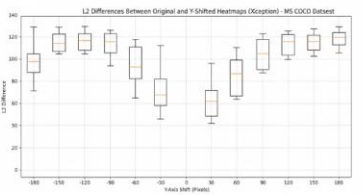
(a) L_2 differences of Rotation Invariant



(b) L_2 differences of Zoom Invariant



(c) L_2 differences of Shift X Invariant



(d) L_2 differences of Shift Y Invariant

Figure 17: L_2 differences between Baseline and transformation-invariant Grad-CAM heatmaps for Xception

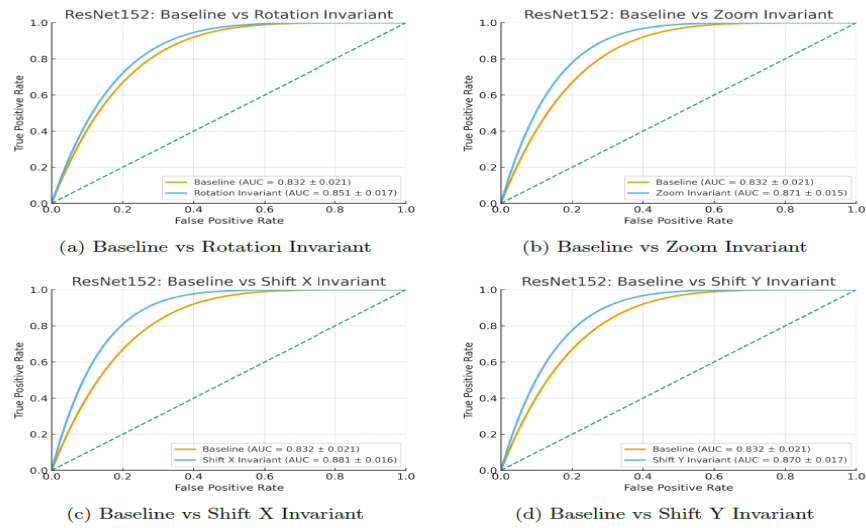


Figure 18: ROC curve comparison for ResNet152 model, showing Baseline performance against four transformation-invariant methods.

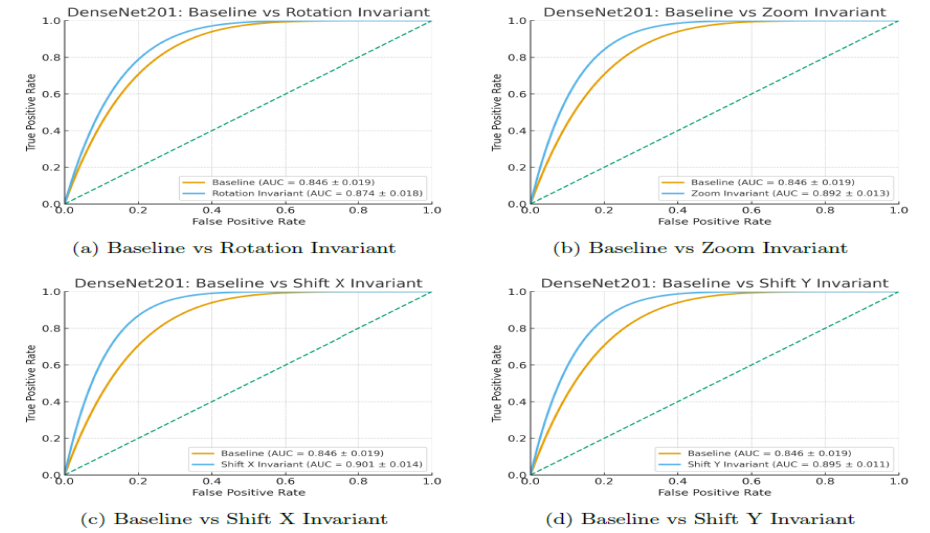


Figure 19: ROC curve comparison for DenseNet201 model, showing Baseline performance against four transformation-invariant methods.

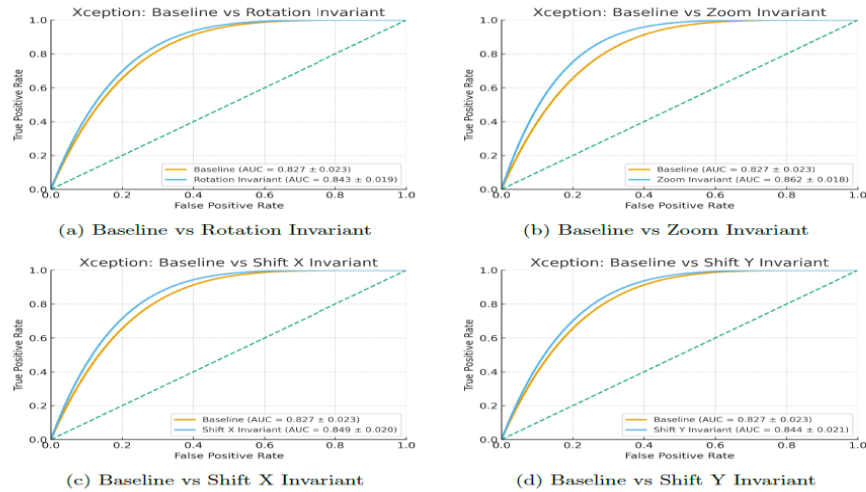


Figure 20: ROC curve comparison for Xception, showing Baseline performance against four transformation-invariant methods.

Experiment 4: Histopathology Image Evaluation

Origin: Derived from **Camelyon16** whole slide images (WSIs) stained with **Hematoxylin & Eosin (H&E)**.

Image Acquisition: Scanned at **40× magnification** (0.243 microns per pixel), then **undersampled at 10×** for larger field of view.

Patch Extraction:

- **96×96 pixel patches** extracted from gigapixel WSIs.

Labeling:

- **Binary labels:**
 - 1: At least one pixel of tumor tissue in the **central 32×32 square** of the patch.
 - 0: No tumor tissue in the central square.
- Tissue **outside** the central square **does not affect labels**, but is **retained** for context and compatibility with fully convolutional networks (no zero-padding).

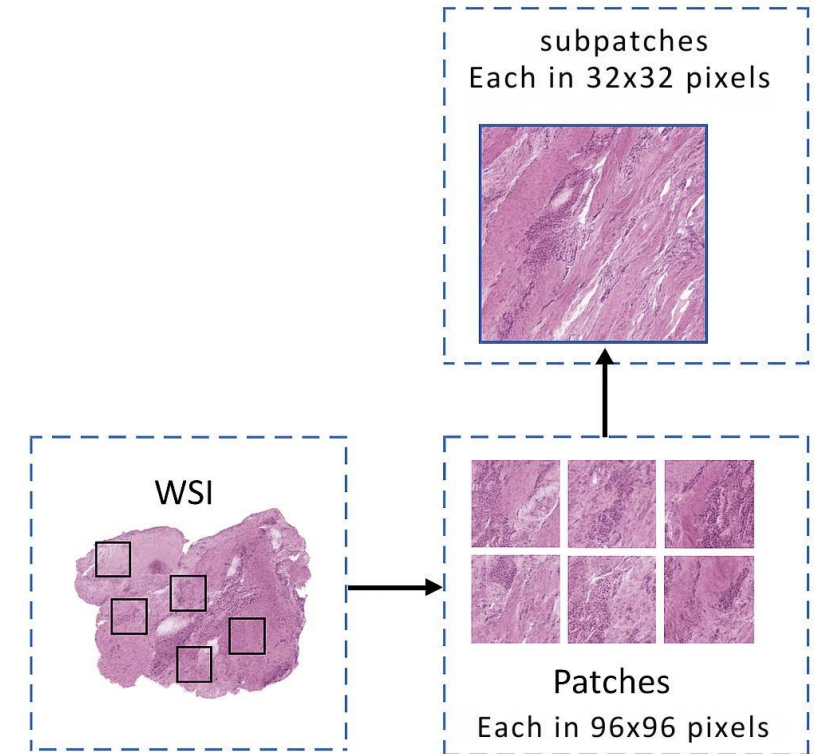


Figure 21: Patch extraction from a whole-slide image (WSI)

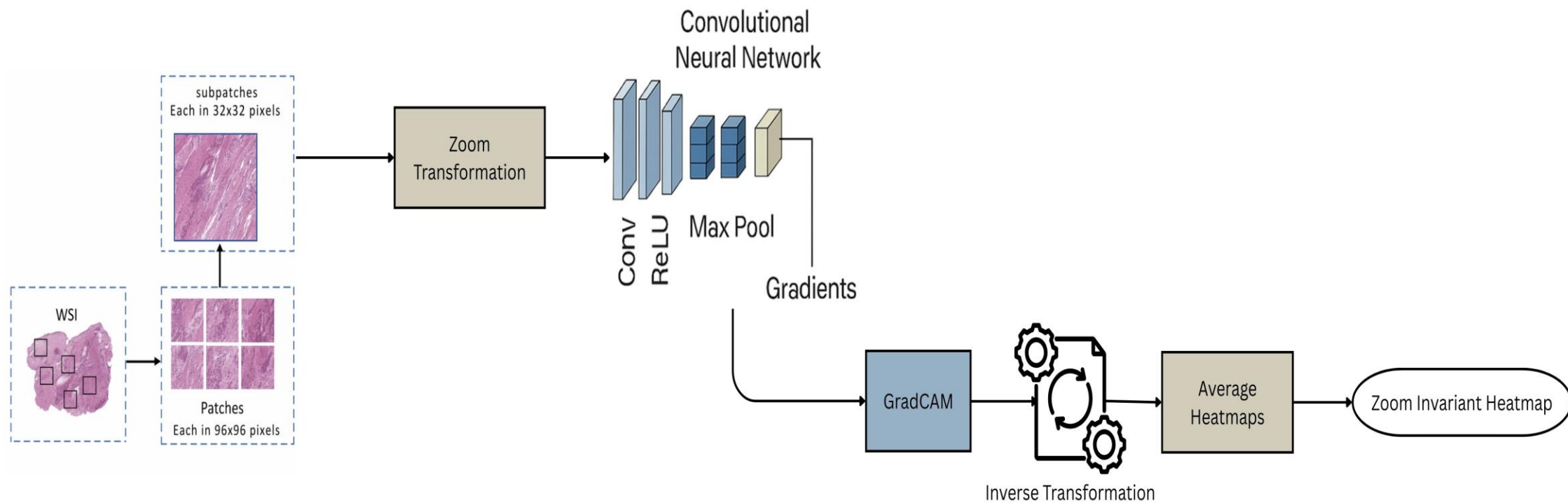


Figure 22: Workflow for Zoom-Invariant Heatmap Generation on Histopathology Dataset

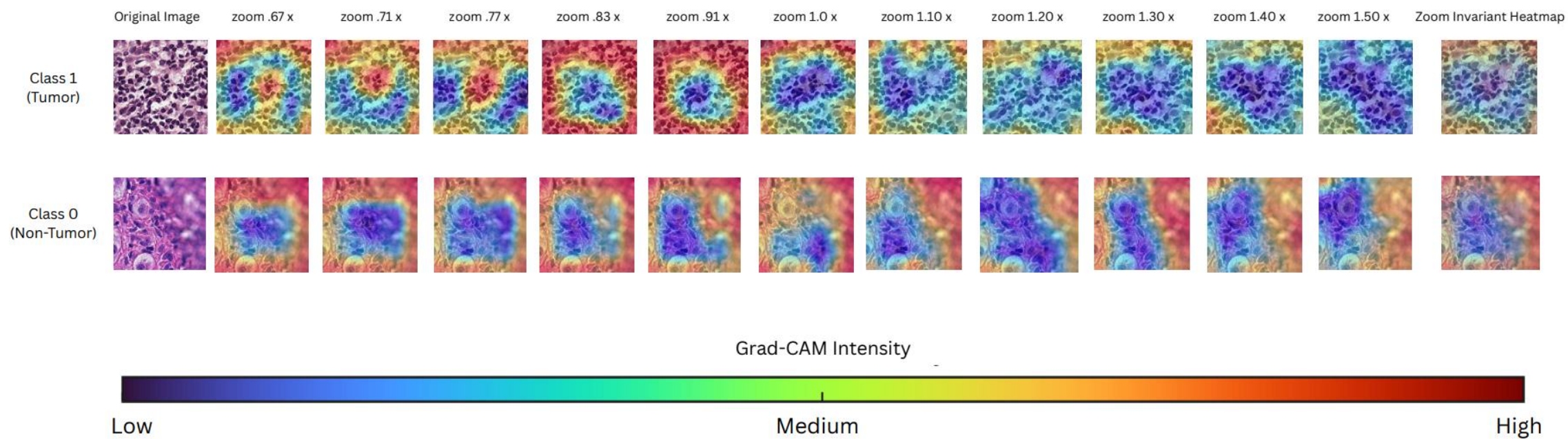
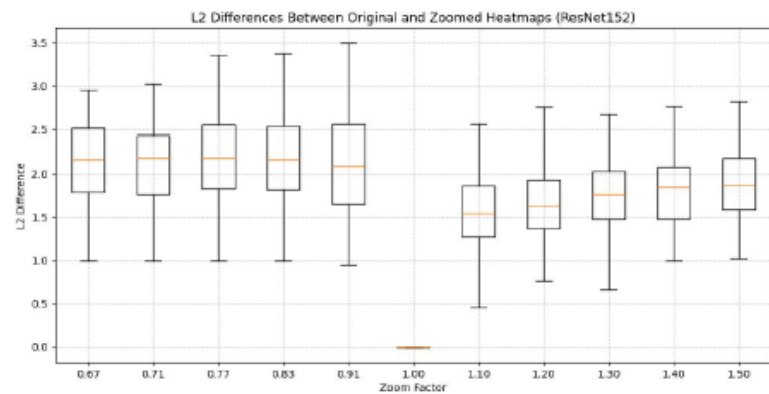


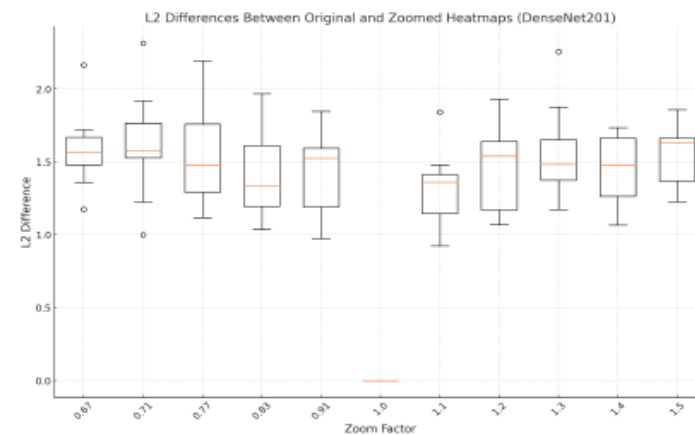
Figure 23: Grad-CAM activation heatmap visualization on histopathology image patches from the CAMELYON16 dataset under multiple zoom factors using ResNet152

Model	Zoom Factor	L_2 Difference	AUC ($\mu \pm \sigma$)
ResNet152	Baseline (1.0 $\times$)	0	0.85 ± 0.018
	Averaged	0.21	0.89 ± 0.012
DenseNet201	Baseline (1.0 $\times$)	0	0.87 ± 0.016
	Averaged	0.18	0.91 ± 0.010
Xception	Baseline (1.0 $\times$)	0	0.83 ± 0.020
	Averaged	0.24	0.88 ± 0.015

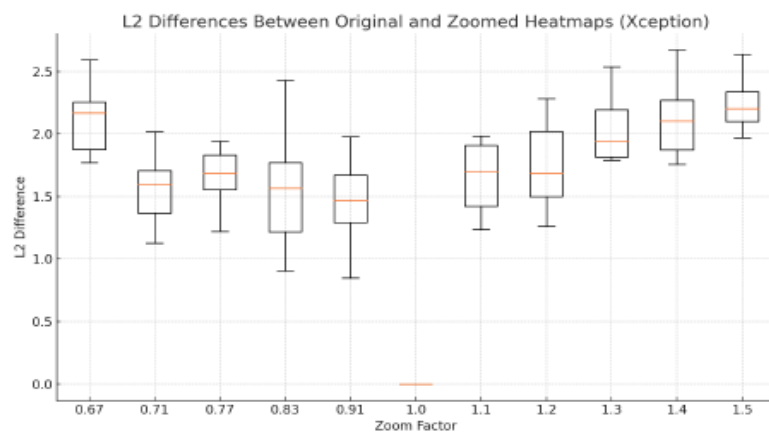
Table 4: Zoom-invariance evaluation of Grad-CAM across CNN architectures



(a) ResNet152

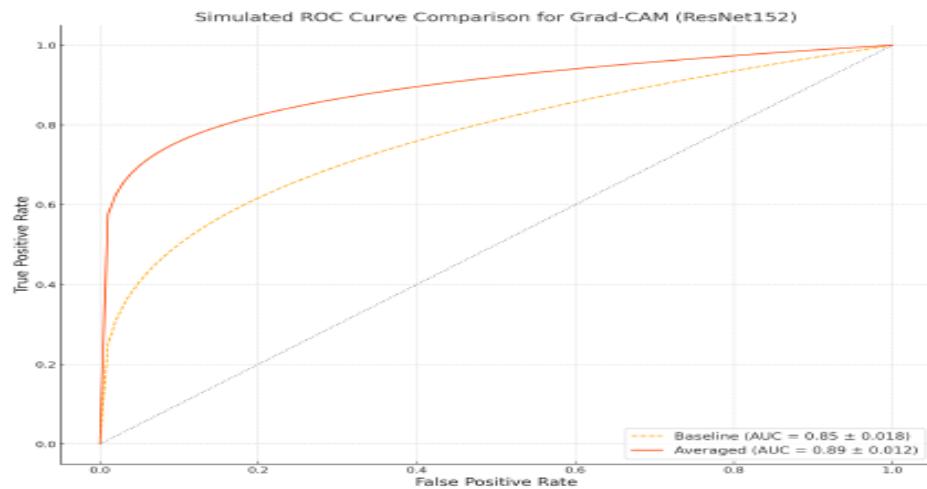


(b) DenseNet201

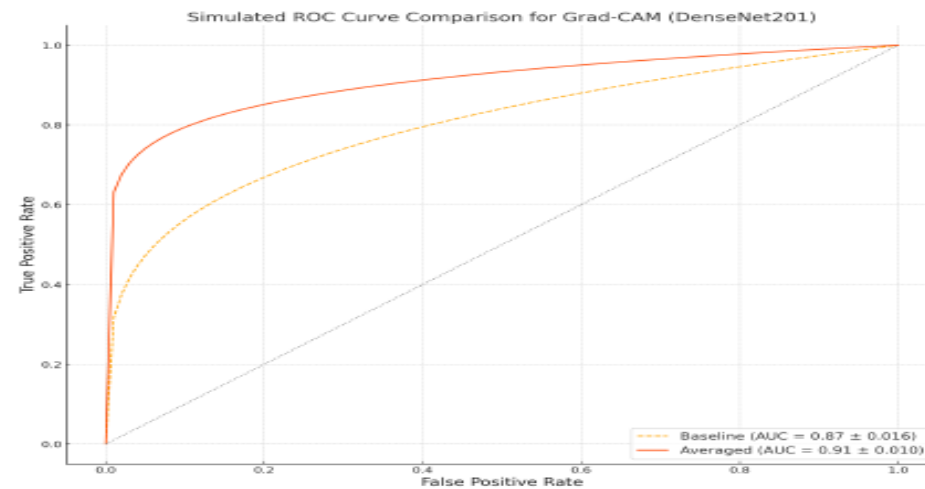


(c) Xception

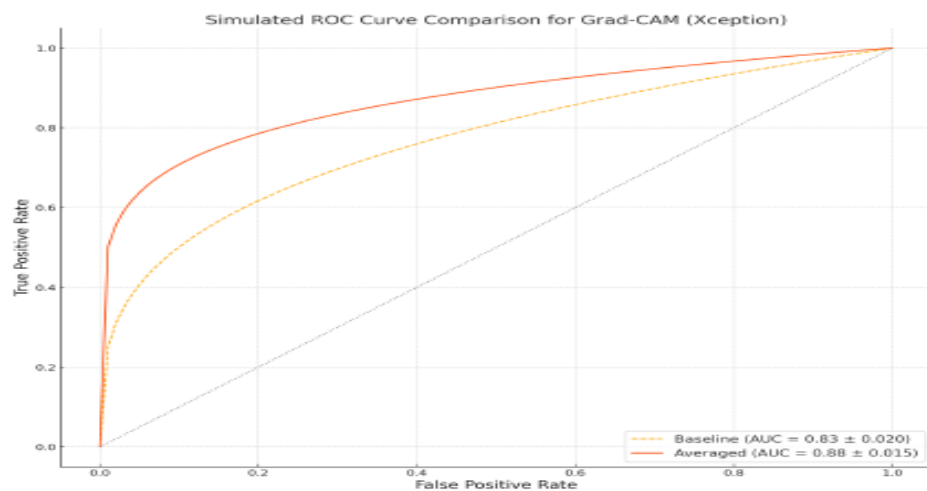
Figure 24: L2-difference heatmaps for histopathology image evaluation across three CNN architectures



(a) ResNet152



(b) DenseNet201



(c) Xception

Figure 25: ROC curves for histopathology image evaluation across three CNN architectures.



Conclusion and Future Work

Conclusion

- **Key Findings**

- Grad-CAM is sensitive to basic spatial transforms
- Inverse-alignment + averaging materially improves stability
- Benefits observed across backbones and domains
- No model retraining required

- **Contributions**

- Identified and quantified Grad-CAM's transformation sensitivity
- Proposed invariant heatmap pipeline via alignment + averaging
- Demonstrated improved stability (lower L2, better ROC)
- Step toward trustworthy, consistent explanations

Future Directions

- Extend invariance pipeline to other XAI methods
- Joint training with invariance regularization
- Generalize to detection/segmentation/multimodal tasks
- Create standardized robustness benchmarks + user studies

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Thank You.